Physics-Based Telemetry Simulator

Technical Documentation — COMP6016

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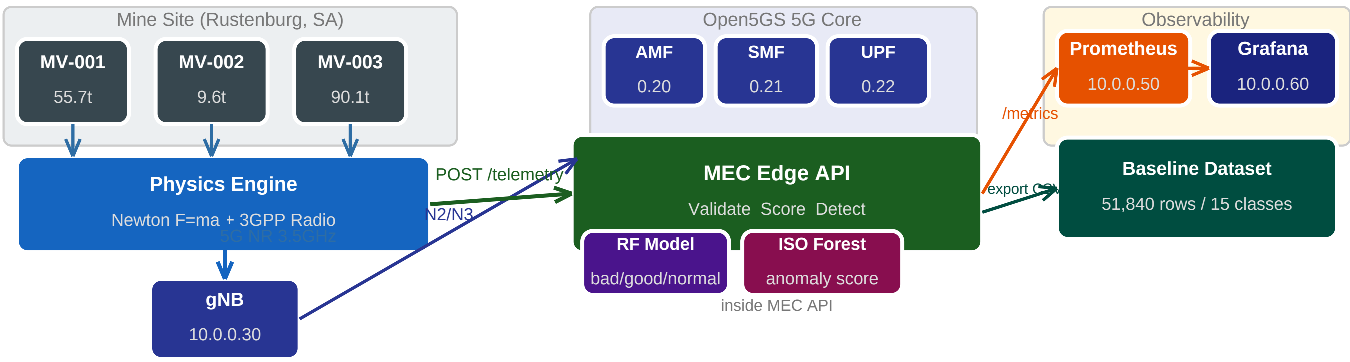
1. What It Does and Why It Exists

The physics simulator is the data engine of the entire project. Its job is to pretend to be three Komatsu 730E haul trucks driving around a platinum mine, and every 5 seconds send a realistic snapshot of each truck's state to the MEC Edge API and Prometheus.

The reason it exists is simple: the previous COMP6015 simulator generated completely random numbers for every metric. Fuel could go up. Temperature had nothing to do with speed. RSRP was unrelated to where the truck was. A machine learning model trained on that data learns noise, not physics. When you then try to detect attacks — which cause specific physical signatures like RTT spikes or packet loss spikes — the model has no baseline to compare against because the "normal" data was never physically realistic to begin with.

This simulator fixes that by modelling the actual physics of the truck and the actual 3GPP radio equations. The result is data where everything is causally linked — when MV-003 climbs a hill with 90 tonnes on board, it automatically slows down, burns more fuel, runs hotter, and its 5G signal degrades, all simultaneously and consistently.

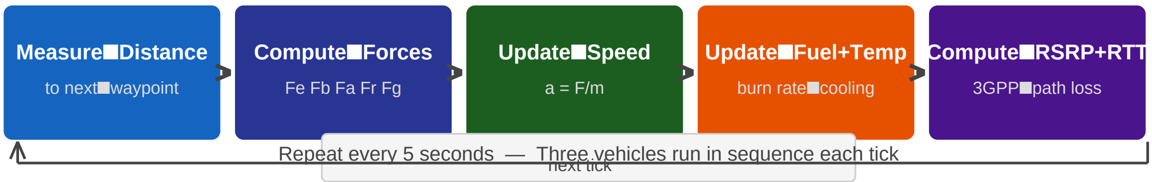
Fig. 1 — Full System Pipeline: The physics engine drives all three vehicles through the haul road circuit. Every 5 seconds it POSTs telemetry to the MEC Edge API and publishes Prometheus metrics. The 5G registration path runs through UERANSIM and the Open5GS core in parallel. Grafana visualises everything in real time.



2. How One Simulation Tick Works

Every 5 seconds the simulator runs one "tick" for each of the three vehicles in sequence. Each tick goes through five stages:

Fig. 2 — One simulation tick: five sequential stages per vehicle, repeated every 5 seconds.



Stage 1: Measure Distance to Next Waypoint

The truck has a current GPS position and a target waypoint. The Euclidean distance in metres is calculated by converting the latitude/longitude difference to metres (1 degree latitude = 111,320 m). If the truck is within 8 metres of the waypoint it advances to the next one.

Stage 2: Compute Forces — Newton's Second Law

Four forces act on the truck simultaneously:

Force	Formula	When Active
Engine Force (Fe)	min(895 kW × 1000 / speed, 200,000 N)	When speed < target speed
Braking Force (Fb)	mass × 1.5 N/kg	When speed > target speed
Aerodynamic Drag (Fa)	0.5 × 1.225 × 0.85 × 15 × v^2	Always, increases with v^2
Rolling Resistance (Fr)	0.02 × mass × 9.81	Always when moving
Grade Force (Fg)	sin(grade°) × mass × 9.81	On slopes (+10% up, –8% down)

Stage 3: Update Speed

Net force divided by mass gives acceleration. New speed = old speed + acceleration × 5 s × 0.4. The 0.4 damping factor prevents unrealistic jerky acceleration. Speed is bounded between 0 and the vehicle's maximum (48 km/h loaded, 56 km/h empty).

Stage 4: Update Fuel and Temperature

Fuel burn rate increases with speed and load:

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burn = (0.003 + 0.025 × speed_ratio × load_ratio) × 5 seconds
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Temperature rises when the engine works hard and cools proportionally to how far above ambient it is:

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heat = speed_ratio × load_ratio × 4.0
cool = 0.015 × (temp − 30°C ambient)
new_temp = temp + (heat − cool) × 5 × 0.08
```

This produces thermal equilibrium — the engine stabilises at a temperature where heat generation matches cooling, exactly like a real diesel engine.

Stage 5: Compute 5G Signal Metrics

RSRP is calculated from GPS distance to the gNB using the 3GPP path loss equation:

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FSPL = 20×log10(distance) + 20×log10(3,500,000,000) − 147.55
```

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RSRP = 43 dBm − FSPL − (18 dB if in tunnel) + N(0, 3.5)
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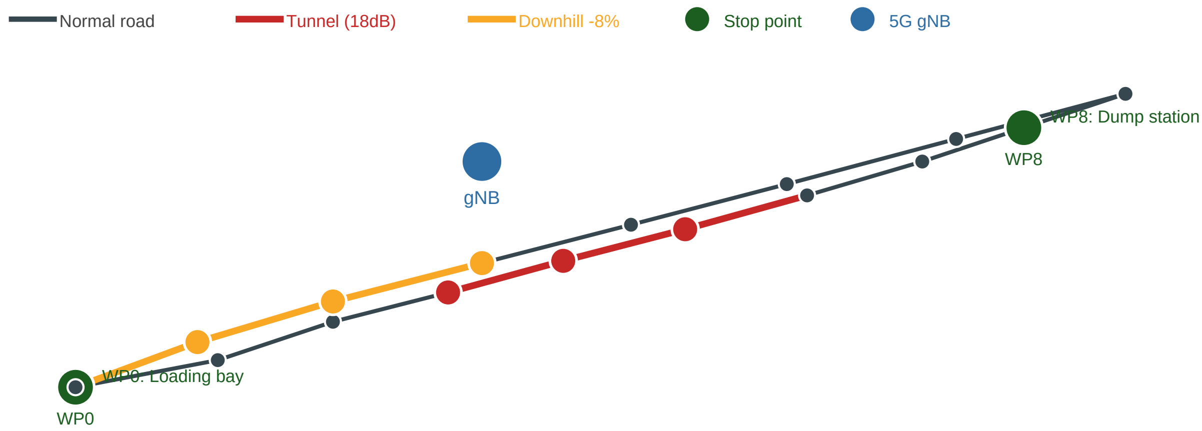
SINR, RTT, packet loss, and throughput are all derived from RSRP, creating a chain of physical causality that runs from GPS position all the way through to network performance.

3. The Haul Road — Real GPS Coordinates

The 17 waypoints are based on the actual Rustenburg Platinum Mine haul road layout in South Africa. The circuit represents a loaded outbound trip followed by an empty return.

Fig. 3 — Haul road circuit: 17 waypoints. Red segments are the underground tunnel (18 dB signal loss). Yellow segments are the downhill return (–8% grade). Green circles are stop points (loading bay WP0 and dump station WP8). Blue circle is the 5G gNB position.

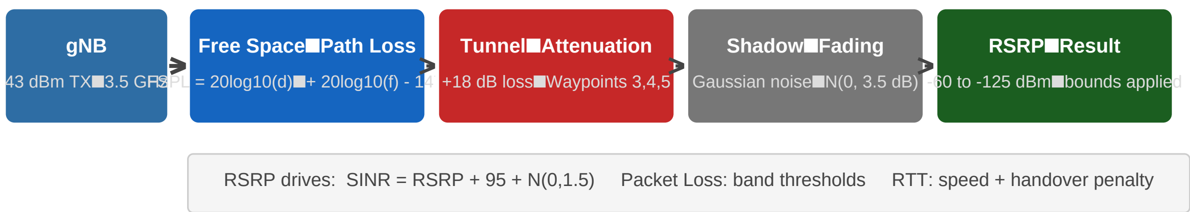


Waypoint(s)	What Happens
WP 0 — Loading bay	Truck stops (4–10 ticks). Takes on 60–180 tonnes random payload. Mass recalculated.
WP 8 — Dump station	Truck stops (4–10 ticks). Payload drops to 0 tonnes. Mass recalculated.
WP 3, 4, 5 — Tunnel	Grade increases to +10%. RSRP drops 18 dB. Packet loss increases significantly.
WP 13, 14, 15 — Downhill	Grade set to –8%. Truck accelerates on return. Engine cooling improves.
All others	Grade is random between –1.5° and +1.5° (minor terrain variation).

4. The 5G Radio Model

The radio model calculates what the 5G signal would actually look like at each truck's GPS position. It uses the 3GPP free-space path loss equation at 3.5 GHz, which is the actual frequency band used in NR band n78 deployments.

Fig. 4 — Radio model chain: GPS distance feeds into 3GPP path loss, then tunnel attenuation and shadow fading are applied to produce RSRP. SINR, packet loss, and RTT are all derived from RSRP.



RSRP Range	5G Quality	Packet Loss	What It Means
Above –70 dBm	Excellent	< 0.1%	Close to gNB, clear line of sight
–70 to –90 dBm	Good	~0.3%	Normal surface operation
–90 to –100 dBm	Fair	~1.5%	Far from gNB or mild obstruction
–100 to –110 dBm	Poor	~5%	Approaching coverage edge
Below –110 dBm	Critical	~15%+	In tunnel or at range limit

5. Scope, Limitations, and Future Work

What is within scope and why it is sufficient

The goal of this project is security testing, not high-fidelity vehicle simulation. Three requirements determine whether the simulator is good enough for this purpose:

- **Normal operation must produce consistent, correlated metric patterns** — satisfied. All four physics validations pass on every run.
- **Attacks must produce clearly distinguishable anomalies** — satisfied. RTT floods, packet loss floods, and impossible sensor values are all detectable.
- **Data must be physically plausible** — satisfied. Real Komatsu 730E specifications and 3GPP path loss equations are used.

Known Limitations

Limitation	Why Not Implemented	Impact on Security Testing
Single gNB	Real mines have 3–5 base stations with inter-cell handover	Low — handover events are still simulated as RSRP threshold crossings
No multi-path fading	Rayleigh fading requires channel model (ITU-R M.2135)	Low — Gaussian noise approximates shadow fading adequately for ML training
No real DEM elevation	Digital Elevation Model requires licensed mine survey data	Low — grade is modelled at key waypoints based on typical haul road design
No V2V interaction	Convoy mode in dataset generator; live sim vehicles are independent	Low — security threats target individual vehicles, not the convoy
Fixed 30°C ambient	Weather API integration not implemented	Very low — thermal patterns are still correctly correlated with speed and load
Simplified battery model	Diesel-electric regeneration is complex	Very low — battery is not a primary security detection feature
No tyre wear	Requires mechanical degradation model	Very low — flat tyre fault exists as a labelled class in dataset generator

What Could Be Added in Future Work

- Multi-cell handover using 3GPP A3 event-triggered procedure with multiple gNBs
- Rayleigh fading channel model for more realistic industrial 5G environments
- Digital Elevation Model integration for accurate grade profiles from real mine survey data
- Weather API integration for ambient temperature and rain effects on signal propagation
- Vehicle-to-vehicle communication for cooperative braking and convoy safety
- Tyre wear model with increasing rolling resistance over a shift
- Multi-mine parameterisation — configurable waypoints for any mine layout

6. Validation — How to Confirm It Is Working Correctly

Run these checks to confirm the physics model is behaving correctly:

Check	Command / Method	Expected Result	What It Proves
Fuel decreasing	Watch fuel_liters in Grafana for 2 min	Bars only go down, never up	Fuel model is correct
Temp tracks speed	Compare speed and temp panels in Grafana	Temp rises when speed rises	Thermal model is correct
MV-003 hottest	Compare engine temp across 3 vehicles	MV-003 (90t load) runs hottest	Load correlation working
RSRP in tunnel	Watch RSRP during waypoints 3–5	RSRP drops ~18 dB in tunnel	Radio model is correct
Prometheus scraping	curl localhost:8000/metrics | grep mining_vehicle_speed	3 gauge values for MV-001/002/003	Metrics exported correctly
MEC receiving data	curl localhost:5000/health	vehicles_seen has 3 entries	POST /telemetry pipeline working